Smart IoT Door Lock with Emergency Alert System

Project Proposal — System Design & Prototype Plan

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Project Type	System Design & Prototype Plan
Domain	IoT, Embedded Systems, Home Security
Technologies	ESP32 / Arduino, Sensors, Wi-Fi, Backend Server, Web/Mobile Interface
Project Status	Concept & Design Phase — Prototype Planned

Acknowledgement

This document was prepared with the assistance of AI tools for writing support, formatting, and background research.

The project idea, problem identification, and proposed system design were conceptualized by me, Vulli Thraiambica. As a beginner exploring IoT systems and system design, I used AI tools to support my learning, improve clarity in documentation, and better understand technical concepts during the research process.

AI tools were used only to organize information, refine explanations, and assist with documentation, while the core concept and intended solution remain entirely my own.

Problem Statement

Traditional door lock systems provide only physical security and do not offer any digital monitoring or alert mechanisms. Homeowners are often unaware of when the door is accessed, especially when away from home.

Key gaps in existing systems include:

- No real-time tracking of lock and unlock events.
- No automated alerts when unauthorized key insertion or forced access is attempted.
- No simple mechanism for occupants to quickly notify nearby people during emergencies.
- Existing smart lock alternatives are often expensive or require complex installation.

> **There is a clear need for a smart, affordable, and easy-to-deploy door lock system that tracks activity in real time, detects unauthorized access, and includes an emergency alert mechanism for occupants.**

Proposed Solution

The Smart IoT Door Lock with Emergency Alert System is an IoT-based residential security solution that monitors door access activity in real time, detects suspicious or unauthorized access attempts, and provides a community-based emergency alert mechanism for occupants.

Core solution components:

- Real-Time Lock Monitoring — Sensors detect every lock and unlock event and report them with timestamps to a web or mobile dashboard.
- Unauthorized Access Detection — The system identifies abnormal key attempts or forced entry and triggers instant alert notifications.
- Audible Buzzer Alert — When unauthorized access or forced entry is detected, a buzzer connected to the system sounds an immediate on-site alarm to warn occupants and deter intruders.
- Emergency Alert Button — A portable remote button allows occupants to immediately notify nearby neighbours and trusted contacts.
- Abort / Cancel — Accidental alerts can be cancelled; neighbours receive a cancellation message to prevent panic.
- Community Escalation — Serious events escalate progressively from neighbours to homeowner to police station.
- Activity Logging — All events stored in a structured log with timestamps for historical review.
- Remote Monitoring Interface — Users can check door status, access history, and alerts from anywhere.

System Architecture

The system is designed using a modular architecture with six integrated components:

Component	Description
Smart Door Lock Unit	Physical lock equipped with sensors to detect lock/unlock actions and unauthorized key insertion or tampering.
Audible Buzzer / Alarm	A buzzer connected to the microcontroller that sounds an immediate on-site alarm when unauthorized access, forced entry, or an emergency button press is detected. Acts as the first physical warning layer.
Microcontroller (ESP32 / Arduino)	Central processing unit that receives sensor data, classifies events, triggers the buzzer, and transmits data to backend via Wi-Fi.
Communication Layer	Wi-Fi connectivity for real-time data transmission from hardware to the cloud backend.
Backend Server / Cloud	Stores all access event logs with timestamps, processes alert conditions, and triggers notifications.
User Monitoring Interface	Web or mobile dashboard showing current lock status, access history, and alert notifications.
Emergency Remote Button	Portable handheld button that triggers instant emergency alert to neighbours, homeowner, and trusted contacts.

System Workflow

Normal Access Workflow

- User locks or unlocks the door using a key.
- Sensors detect the action and send data to the microcontroller.
- Microcontroller classifies the event as normal access.
- Event data is transmitted via Wi-Fi to the backend server.
- Server logs the event with timestamp and updates the monitoring dashboard.

Unauthorized Access Detection Workflow

- Sensors detect an irregular key insertion or forced access attempt.
- Microcontroller flags the event as suspicious activity.
- Buzzer sounds immediately on-site — audible alarm activates to warn occupants and deter the intruder.
- Backend server receives alert data and generates an instant notification.
- Homeowner is alerted in real time; event is recorded in the activity log.

Emergency Alert Workflow

- Occupant presses the portable emergency remote button.
- Emergency signal is sent to the microcontroller → backend server triggered.
- Buzzer sounds immediately on-site as an audible emergency alarm.
- System sends Level 1 alert to nearby registered neighbours immediately.
- If the alert was accidental, the Abort button cancels it — neighbours receive a cancellation message.
- If the situation is unresolved, Level 2 alert is sent to the homeowner and trusted contacts.
- If forced entry or serious threat is confirmed, Level 3 escalation alerts the police station with location details.
- All emergency events are stored in the system activity log.

System Workflow Diagram

The diagram below illustrates the complete system workflow — covering the door access monitoring flow (left) and the emergency alert escalation flow with community response levels (right).

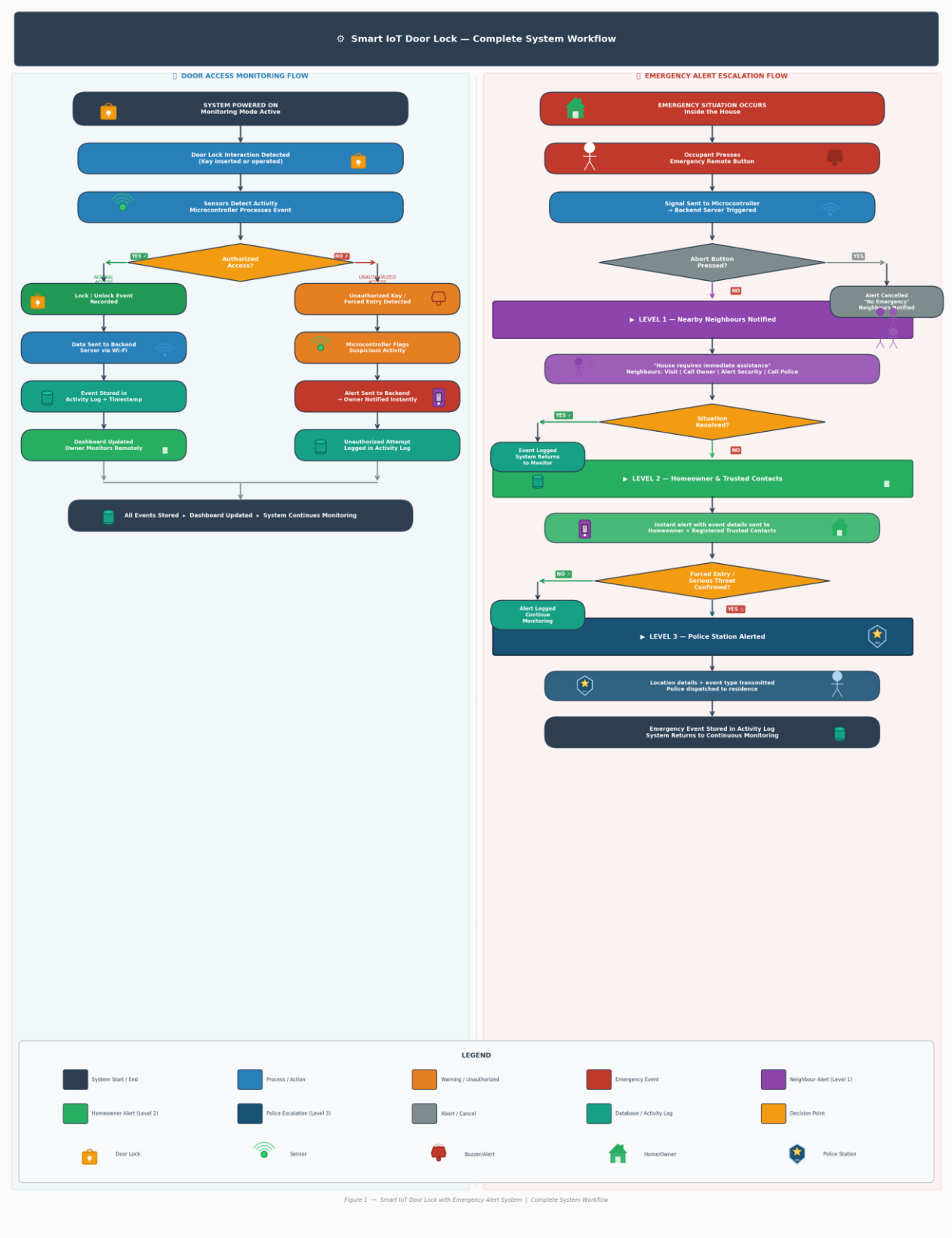

Figure 1 — Complete System Workflow (Door Access + Emergency Alert Escalation)

Community-Based Alert System

A key design principle of this system is a multi-level community alert hierarchy. Rather than depending on a single contact, the system escalates progressively based on the severity of the situation.

Level 1 — Neighbour Alert

When an emergency button is pressed or a serious event is detected, the first alert is sent to registered nearby neighbours. Neighbours are physically closer than emergency services and can respond within minutes.

Possible neighbour actions:

- Physically check on the house or resident.
- Contact the homeowner directly.
- Call emergency services if the situation is confirmed serious.
- Alert local security personnel.

If the alert was accidental, the occupant can press the Abort / Cancel button. Neighbours receive a cancellation notification — preventing unnecessary panic or false response.

Level 2 — Homeowner & Trusted Contacts

Simultaneously or after neighbour notification, the homeowner and registered trusted contacts receive an instant alert with event details, allowing them to take appropriate action remotely.

Level 3 — Police Station Escalation

Police are involved only when the situation is genuinely serious. Scenarios that trigger police escalation:

- Confirmed forced entry detected by the system.
- A serious emergency that neighbours cannot resolve.
- A situation verified as a real threat by neighbours or homeowner.

When escalation triggers, the system sends an automated alert to the local police station containing the location details and event description, enabling a fast and informed response.

Level	Recipient	Trigger Condition
Level 1	Nearby Neighbours	Emergency button pressed or suspicious activity detected.
Level 2	Homeowner / Trusted Contacts	Any emergency or confirmed access violation event.
Level 3	Police Station	Confirmed forced entry, serious unresolved emergency, or verified threat.

Prototype Implementation Plan

The prototype will be developed across five structured phases:

Phase	Title	Description
Phase 1	**Hardware Setup**	Install ESP32/Arduino microcontroller, connect lock/unlock sensors, configure emergency remote button, enable Wi-Fi connectivity.
Phase 2	**Event Detection**	Program the microcontroller to detect normal access, unauthorized access attempts, and emergency button triggers.
Phase 3	**Backend & Logging**	Develop a backend server to receive and store event data with lock status, timestamps, and event types.
Phase 4	**Monitoring Interface**	Build a web or mobile dashboard to view current lock status, access history, and alert notifications remotely.
Phase 5	**Alert Integration**	Integrate real-time notification delivery for neighbour, homeowner, and police escalation alerts.

Key Features

Feature	Description
Real-Time Door Lock Monitoring	Detects and records every lock and unlock event with timestamps.
Timestamped Activity Logging	All events stored in a database for historical review and security awareness.
Unauthorized Access Detection	Identifies irregular key insertions or forced access attempts and triggers instant notifications.
Audible Buzzer / On-Site Alarm	A buzzer sounds an immediate on-site audible alarm when unauthorized access, forced entry, or emergency button press is detected — acting as the first physical deterrent before any digital notification is sent.
Emergency Alert Button	Portable remote button for occupants to instantly trigger emergency notifications.
Abort / Cancel Alert	Cancels accidental alerts; neighbours receive a cancellation message to prevent false panic.
Neighbour Alert System	First-responder alerts sent to nearby registered neighbours — physically closer, faster response.
Homeowner / Contact Alert	Real-time alerts delivered to homeowner and trusted contacts on any serious event.
Police Station Escalation	Automated alert with location details sent to the police station on confirmed forced entry or serious threat.
Remote Monitoring Interface	Web or mobile dashboard for remote access to lock status, activity logs, and alerts.

Technologies Planned

Technology	Role in System
Internet of Things (IoT)	Core architecture enabling device-to-cloud communication and real-time monitoring.
ESP32 / Arduino	Microcontroller acting as the central processing and communication unit.
Sensors	Detect lock/unlock actions and identify unauthorized access attempts.
Buzzer / Alarm Module	Produces an audible on-site alarm signal triggered by the microcontroller on unauthorized access, forced entry, or emergency events.
Wi-Fi Communication	Transmits event data in real time from hardware to the backend server.
Backend Server & Database	Stores activity logs, processes alert conditions, and manages notifications.
Web / Mobile Interface	Dashboard providing remote access to lock status, activity history, and alerts.
Notification System	Delivers real-time alerts to neighbours, homeowner, trusted contacts, and police.

Conclusion

The Smart IoT Door Lock with Emergency Alert System is a concept-stage IoT project designed to address real gaps in residential security. By integrating real-time monitoring, unauthorized access detection, and a community-based emergency alert hierarchy, it demonstrates how IoT technology can be applied practically to improve home safety.

The design, system architecture, workflow logic, and prototype implementation plan in this document reflect my independent thinking about the problem and approach to solving it. The core idea — including the emergency button concept, neighbour-first alert mechanism, abort functionality, and police escalation — was conceived entirely by me.

I am eager to explore this concept further and build on it under the guidance of researchers and mentors at IIT Hyderabad. I see this internship as an opportunity to strengthen my technical foundation in IoT and embedded systems and bring this project from design to a working prototype.